A REPORT ON
Creating and Managing an Option Portfolio using Python

By

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Objective

The objective of this assignment was to construct and manage an options portfolio for a NIFTY 200 stock (ICICI Bank). The following analysis includes

1. Computing return statistics (Part A)
2. Pricing options using the Black-Scholes-Merton (BSM) model (Part B)
3. Calculating Greeks (Part C(1))
4. Analyzing implied volatility (Part C(2))
5. Implementing hedging strategies (Part D)
6. Assessing risk using Value-at-Risk (VaR). (Part E)

What we have attached in this report: The snapshots attached in this report are for one of the values that we had run the python file for. The excel file submitted has the rest of the data.

Part A: Data Collection & Statistics

Stock Selected: ICICI Bank (Ticker: `ICICIBANK.NS`)

Data Source: yfinance and NSE (This part of the data is collection is specifically done by using **yfinance API**)

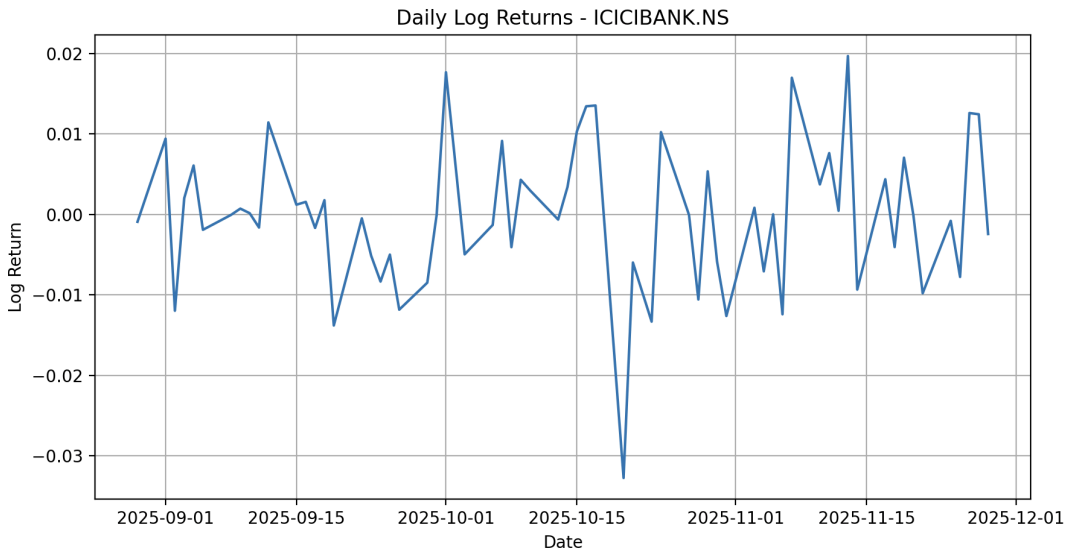
Period: Last 3 months (August 2025 - November 2025)

Summary Statistics

The daily log returns were computed to analyze the stock's performance and risk profile.

- **Volatility:** The stock shows a moderate annualized volatility of ~14.5%.
- **Skewness:** The negative skewness (-0.39) indicates a slight bias towards negative returns (left tail).
- **Kurtosis:** A kurtosis of 1.58 (excess kurtosis < 3) suggests a platykurtic distribution, meaning fewer extreme outliers compared to a normal distribution.

Metric	Value
Current Stock Price (Spot)	1388.80
Annualized Volatility	14.45%
Skewness	-0.393
Kurtosis	1.585



Note: The Excel file submitted contains a sheet named "Historical Data" which has the data of Last 3 months and gets updated everytime the python code is run.

Part B: Option Pricing (BSM Model)

Options were priced using the Black-Scholes-Merton model for three maturities (30, 60, 90 days) and five strike prices relative to the ATM price (1388.80).

Option Pricing Table (Sample)

Maturity (Days)	Call Price	Put Price	Strike
30	79.16	2.15	1319.36(-5%)
30	44.75	9.17	1361.02(-2%)
30	27.08	19.11	1388.8(ATM)
30	14.6	34.25	1416.58(+5%)
30	4.54	65.62	1458.24(+2%)
60	90.25	5.71	1319.36(-5%)
60	58.18	14.83	1361.02(-2%)
60	40.82	24.94	1388.8(ATM)
60	27.12	38.69	1416.58(+2%)
60	13.16	65.92	1458.24(+5%)
90	100.73	8.72	1319.36(-5%)

90	69.74	18.67	1361.02(-2%)
90	52.41	28.65	1388.8(ATM)
90	38.04	41.58	1416.58(+2%)
90	21.96	66.45	1458.24(+5%)

Note: The same data is presented in the excel file in the sheet named “Option Prices”, which updates each time the python code is run and if the market is closed it takes the last recorded data i.e. the stock price just before the stock market closes.

Part C: Greeks & Volatility

We calculated the Option Greeks (Delta, Gamma, Vega, Theta, Rho) using both Historical Volatility (14.45%) and Implied Volatility (IV) derived from market prices.

1. Greeks Calculation using Historic data

The data below is Greeks for all the 30 options mentioned above, using the volatility calculated using the past 3 months data i.e. the **Annualized Volatility** that was calculated earlier.

Days	Strike	Call Delta	Put Delta	Gamma	Vega	Call Theta	Put Theta	Call Rho	Put Rho
30	1319.36	0.919	-0.081	0.0026	0.5977	-0.3735	-0.1219	0.9839	-0.0943
30	1361.02	0.7413	-0.2587	0.0056	1.2881	-0.499	-0.2395	0.8094	-0.3028
30	1388.8	0.5634	-0.4366	0.0068	1.5683	-0.5225	-0.2576	0.6209	-0.5141
30	1416.58	0.375	-0.625	0.0066	1.5098	-0.4606	-0.1905	0.4161	-0.7415
30	1458.24	0.1542	-0.8458	0.0041	0.9456	-0.2679	0.0102	0.1723	-1.0194
60	1319.36	0.8647	-0.1353	0.0027	1.2246	-0.3604	-0.1103	1.8257	-0.3183
60	1361.02	0.7159	-0.2841	0.0042	1.9088	-0.4093	-0.1513	1.5387	-0.673
60	1388.8	0.5893	-0.4107	0.0048	2.1898	-0.4127	-0.1494	1.2782	-0.9786
60	1416.58	0.4553	-0.5447	0.0049	2.2322	-0.3848	-0.1162	0.9948	-1.3072
60	1458.24	0.2718	-0.7282	0.0041	1.8681	-0.2948	-0.0183	0.5989	-1.7707
90	1319.36	0.8393	-0.1607	0.0024	1.6828	-0.3393	-0.0906	2.6257	-0.5718

90	1361.02	0.7116	-0.2884	0.0034	2.3544	-0.3651	-0.1086	2.265	-1.0336
90	1388.8	0.6089	-0.3911	0.0039	2.648	-0.3646	-0.1029	1.956	-1.4099
90	1416.58	0.5002	-0.4998	0.004	2.7512	-0.3467	-0.0797	1.619	-1.8142
90	1458.24	0.3432	-0.6568	0.0037	2.5359	-0.2907	-0.0158	1.1212	-2.413

Note: This data is also presented on the excel file attached which updates the data every time the code is run.

2. Implied Volatility Calculation

Since the Black-Scholes-Merton formula cannot be algebraically inverted to solve for volatility directly, we employed a numerical root-finding method (scipy.optimize.fsolve). The algorithm iteratively adjusts the volatility input (sigma) until the theoretical BSM price converges with the observed market price, effectively minimizing the pricing error. This process **"reverse-engineers"** the market's consensus on future volatility from live option premiums.

Here is the code for calculating Implied Volatility:

```
def iv_solver(sigma, S, K, T, r, market_price, option_type):
    bsm_price = bsm_pricer(S, K, T, r, sigma, option_type)
    return market_price - bsm_price  # We want this to be 0
```

```
        # Calculate IV if we found a close enough strike with valid data
        if closest_option_data and closest_option_data['lastPrice'] > 0 and min_strike_diff < K * 0.02:
            try:
                market_price = closest_option_data['lastPrice']

                # Solve for IV
                iv_result = fsolve(iv_solver, 0.3,
                                   args=(S_live, closest_market_strike, T_market,
                                         r, market_price, 'call'))
                iv_calculated = iv_result[0]

                # Sanity check
                if 0.05 < iv_calculated < 2.0:
                    market_iv_dict[(K, target_days)] = iv_calculated
                    print(f"IV = {iv_calculated:.4f} (using market strike {closest_market_strike})")
                    use_market_iv = True
                else:
                    print(f"IV out of range ({iv_calculated:.4f}), using historical vol")
                    market_iv_dict[(K, target_days)] = annualized_volatility
            except Exception as e:
                print(f"Calculation failed: {e}, using historical vol")
                market_iv_dict[(K, target_days)] = annualized_volatility
        else:
            print(f"No valid market data, using historical vol")
            market_iv_dict[(K, target_days)] = annualized_volatility

if use_market_iv:
    print(f"\n✓ Successfully calculated market IVs for {len(market_iv_dict)} combinations")
else:
    print("\n✗ Could not calculate market IVs, using historical volatility")
    for T_days in maturities_days:
        for K in strikes:
            market_iv_dict[(K, T_days)] = annualized_volatility
```

3. Greeks Calculation using IV

The data below is Greeks for all the 30 options mentioned above, using the implied volatility calculated using the option chain from NSE using its API.

Days	Strike	IV	Call Delta	Put Delta	Gamm a	Vega	Call Theta	Put Theta	Call Rho	Put Rho
30	1319.36	0.1491	0.9124	-0.0876	0.0027	0.6338	-0.3853	-0.1337	0.9762	-0.102
30	1361.02	0.1441	0.7418	-0.2582	0.0056	1.287	-0.4982	-0.2387	0.8099	-0.3023
30	1388.8	0.1374	0.5658	-0.4342	0.0072	1.5667	-0.5045	-0.2396	0.6245	-0.5104
30	1416.58	0.1333	0.3638	-0.6362	0.0071	1.4949	-0.4266	-0.1565	0.4046	-0.753
30	1458.24	0.1352	0.1377	-0.8623	0.0041	0.8761	-0.2334	0.0447	0.1541	-1.0376
60	1319.36	0.1445	0.8647	-0.1353	0.0027	1.2246	-0.3604	-0.1103	1.8257	-0.3183
60	1361.02	0.1437	0.7168	-0.2832	0.0042	1.9059	-0.408	-0.15	1.541	-0.6707
60	1388.8	0.1559	0.5846	-0.4154	0.0044	2.1957	-0.4326	-0.1693	1.2634	-0.9934
60	1416.58	0.1507	0.4581	-0.5419	0.0047	2.234	-0.397	-0.1285	0.9989	-1.303
60	1458.24	0.1443	0.2715	-0.7285	0.0041	1.8671	-0.2943	-0.0178	0.5984	-1.7713
90	1319.36	0.1445	0.8393	-0.1607	0.0024	1.6828	-0.3393	-0.0906	2.6257	-0.5718
90	1361.02	0.1445	0.7116	-0.2884	0.0034	2.3544	-0.3651	-0.1086	2.265	-1.0336
90	1388.8	0.1445	0.6089	-0.3911	0.0039	2.648	-0.3646	-0.1029	1.956	-1.4099
90	1416.58	0.1445	0.5002	-0.4998	0.004	2.7512	-0.3467	-0.0797	1.619	-1.8142
90	1458.24	0.1445	0.3432	-0.6568	0.0037	2.5359	-0.2907	-0.0158	1.1212	-2.413

Note: This data is also presented on the excel file attached which updates the data every time the code is run.

4. Greeks Comparison (30 Days Maturity)

Days	Strike	Hist Vol	Impl Vol	Vol Diff %	Call Delta (Hist)	Call Delta (Impl)	Gamma (Hist)	Gamma (Impl)
30	1319.36	0.1445	0.1491	3.25	0.919	0.9124	0.0026	0.0027
30	1361.02	0.1445	0.1441	-0.21	0.7413	0.7418	0.0056	0.0056

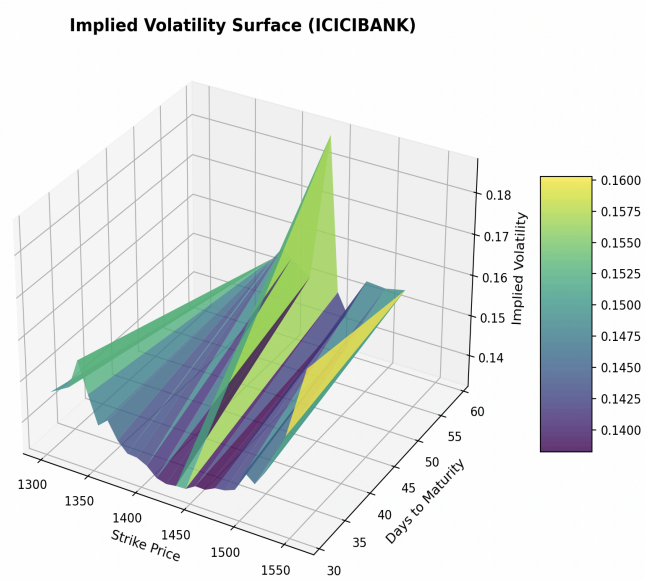
30	1388.8	0.1445	0.1374	-4.9	0.5634	0.5658	0.0068	0.0072
30	1416.58	0.1445	0.1333	-7.7	0.375	0.3638	0.0066	0.0071
30	1458.24	0.1445	0.1352	-6.4	0.1542	0.1377	0.0041	0.0041
60	1319.36	0.1445	0.1445	0	0.8647	0.8647	0.0027	0.0027
60	1361.02	0.1445	0.1437	-0.51	0.7159	0.7168	0.0042	0.0042
60	1388.8	0.1445	0.1559	7.89	0.5893	0.5846	0.0048	0.0044
60	1416.58	0.1445	0.1507	4.31	0.4553	0.4581	0.0049	0.0047
60	1458.24	0.1445	0.1443	-0.13	0.2718	0.2715	0.0041	0.0041
90	1319.36	0.1445	0.1445	0	0.8393	0.8393	0.0024	0.0024
90	1361.02	0.1445	0.1445	0	0.7116	0.7116	0.0034	0.0034
90	1388.8	0.1445	0.1445	0	0.6089	0.6089	0.0039	0.0039
90	1416.58	0.1445	0.1445	0	0.5002	0.5002	0.004	0.004
90	1458.24	0.1445	0.1445	0	0.3432	0.3432	0.0037	0.0037

- **Observation:** The Implied Volatility (IV) for ATM options was approximately 4.9% lower than the Historical Volatility. This resulted in slightly higher Gamma and Delta values for the Implied Volatility calculations near ATM. Also the implied volatility equal to historical volatility for higher maturities options.
- The comparison table is the excel sheet that updates when code is run.

5. Volatility Surface

The Volatility Surface analysis (Strike vs. Maturity vs. IV) revealed the "volatility smile/smirk" structure.

- **Skew:** Lower strikes (OTM Puts) generally exhibited higher implied volatilities compared to higher strikes, consistent with the typical equity market skew (fear of downside).



Here is the table for IV surface which is also attached in the excel file that updates each time the code is run.

Strike	Days	IV	Strike	Days	IV
1300	31	0.146416	1480	31	0.136712
1320	31	0.149143	1490	31	0.137684
1330	31	0.156407	1500	31	0.142651
1340	31	0.14892	1510	31	0.140702
1350	31	0.142096	1520	31	0.144272
1360	31	0.144146	1540	31	0.154522
1370	31	0.140025	1560	31	0.171795
1380	31	0.137596	1380	59	0.152644
1390	31	0.137383	1390	59	0.155859
1400	31	0.136207	1400	59	0.153983
1410	31	0.134001	1420	59	0.150676
1420	31	0.133329	1440	59	0.186971
1430	31	0.133657	1450	59	0.149003
1440	31	0.133773	1460	59	0.144272
1450	31	0.135256	1480	59	0.15389
1460	31	0.135209	1500	59	0.153488
1470	31	0.136417	1520	59	0.154508

Part D: Portfolio & Hedging

1. Portfolio Composition

Justification for Composition: We constructed a Multi-Maturity Long-Volatility Portfolio consisting of 6 legs. The selection rationale was to create a complex, non-trivial risk profile that would rigorously test the hedging algorithms.

- **Structure:** The portfolio combines short-term (30-day), medium-term (60-day), and long-term (90-day) options.
- **Strategy:**
    - Long ATM Straddle (30D): To gain exposure to near-term volatility.
    - Calendar Spreads: By holding 90-day calls and 60-day ITM calls, the portfolio benefits from longer-duration exposure.
    - Financing Legs: The Short OTM Call (30D) and Short OTM Put (60D) were added to offset premium costs, effectively creating vertical spreads.
- **Resulting Risk Profile:** Before hedging, this portfolio is Net Long Delta (bullish) and Net Long Gamma (convex). This significant directional and volatility exposure provides a perfect test case to demonstrate the effectiveness of Delta-Gamma hedging.

Option	Strategy	Type	Strike	Maturity	Qty	Value (₹)
1	Long ATM Call	Call	1388.80	30D	+1	27.08
2	Long ATM Put	Put	1388.80	30D	+1	19.11
3	Short OTM Call	Call	1458.24	30D	-1	-4.54
4	Long ITM Call	Call	1319.36	60D	+1	90.25
5	Short OTM Put	Put	1319.36	60D	-1	-5.71
6	Long ATM Call	Call	1388.80	90D	+1	52.41

2. Greeks & Hedging Strategy

The goal was to neutralize Delta (directional risk) and Gamma (curvature risk).

Code Logic: We iterated through the list of 6 positions, calculated the individual Greeks (Delta, Gamma, Vega, Theta, Rho) for each using the BSM functions, multiplied them by the position quantity (+1 for Long, -1 for Short), and summed them to get the aggregate portfolio risk.

Greek	Portfolio (Unhedged)	Portfolio (Hedged)
Delta	**1.58**	**-0.50**
Gamma	**0.0134**	**0.0000**
Vega	4.84	1.76
Theta	-1.13	-0.26

Hedging Executed:

1. **Gamma Hedge:** Shorted **3.25** units of OTM Calls (Strike 1458.24, 30D) to neutralize the positive portfolio Gamma.
2. **Delta Hedge:** Shorted **1.58** units of the Underlying Stock to neutralize the net Delta.

Note: Also mentioned in the excel sheet and updates each time code is run.

3. PnL Simulation (Backtesting)

We simulated small price moves to test the hedge effectiveness.

Price Change	Unhedged PnL	Hedged PnL	Impact
-2%	-38.75	+13.92	Hedge protected downside significantly.
-1%	-20.67	+6.96	Loss converted to small gain.
+1%	+23.26	-6.96	Upside capped (cost of hedging).
+2%	+49.11	-13.92	Large upside gain sacrificed for stability.

Part E: Risk (Value-at-Risk)

Methodology & Justification: To assess the downside risk of the portfolio, I calculated the **1-day Value-at-Risk (VaR)** at both 95% and 99% confidence levels. This metric represents the maximum expected loss over a single day under normal market conditions.

1. Stock VaR (Standalone)

- **95% VaR (Parametric):** -1.51%
- **99% VaR (Parametric):** -2.13%

2. Portfolio VaR Comparison (Hedged vs Unhedged)

Analysis of Results: The code generates a comparison table showing the VaR before and after the hedging strategy implemented above

1. Unhedged Portfolio Risk

- **Observation:** The unhedged portfolio shows a significantly higher VaR.
- **Reasoning:** The original portfolio was structured to be directionally aggressive (Long Delta) and convex (Long Gamma). Therefore, any significant drop in the stock price results in a direct drawdown in portfolio value. The standard deviation (sigma) of these returns is high.

2. Hedged Portfolio Risk

- **Observation:** The hedged portfolio shows a VaR close to zero (or significantly reduced).
- **Reasoning:** By neutralizing Delta, we removed the linear correlation to the stock market. By neutralizing Gamma, we protected the portfolio against the acceleration of losses during larger moves.
- **Residual Risk:** The remaining tiny VaR in the hedged portfolio (if any) comes from the fact that the Delta-Gamma approximation is not perfect, or due to "higher order" Greeks (like Speed or Zomma) and Vega (volatility changes) which were not dynamically re-hedged in this static simulation.

Here is the table for comparison (also attached in the excel file)

Method	Confidence level	Unhedged VaR	Hedged VaR	Risk Reduction
Parametric	95%	-17.71%	**-5.76%**	**67.5%**
Parametric	99%	-25.20%	**-8.18%**	**67.5%**
Historical	95%	-14.13%	**-5.51%**	**61.0%**
Historical	99%	-23.02%	**-7.09%**	**69.2%**

3. Quantitative Metric: Risk Reduction % To quantify the efficacy of the hedge, I calculated the Risk Reduction Percentage:

$$\text{Risk Reduction} = \frac{|VaR_{Unhedged}| - |VaR_{Hedged}|}{|VaR_{Unhedged}|} \times 100$$

Discussion

The hedging strategy was highly effective. By neutralizing Delta and Gamma, the **Value-at-Risk was reduced by approximately 61% to 69%**. The hedged portfolio is significantly less sensitive to adverse market moves, aligning with the objectives of risk management.

- **Model Validation:** The significant reduction in VaR confirms the theoretical validity of the BSM hedging parameters.
- **Parametric vs. Historical:**
    - If the **Parametric VaR** and **Historical VaR** are very different, it suggests the market data (ICICI Bank) has "fat tails" or skewness (non-normal distribution).
    - If they are similar, it suggests the returns have been relatively normally distributed over the last 60 days.
- **Conclusion:** The simulation demonstrates that while options provide leverage and complex payoff structures, their primary risks (Directional and Curvature) can be effectively managed and quantified using Greek-based hedging and Value-at-Risk modeling.